

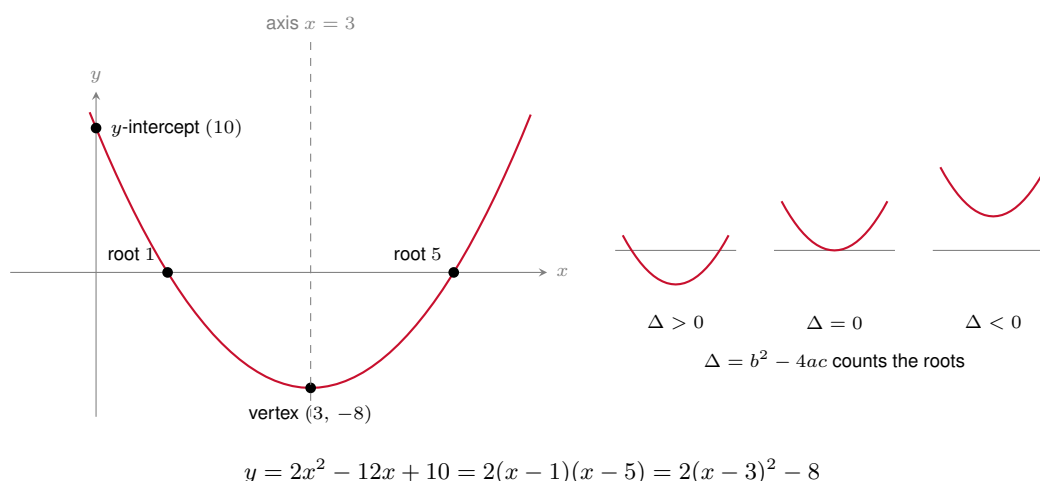
Quadratics: one curve, three forms, and VAR

THE ONE IDEA

A quadratic is one curve in three forms. Each form hands you one fact for free. The cost is expanding the brackets.

Board order: $y = x^2$ (domain $\mathbb{R}$, range $y \geq 0$), then the sign of a and Δ , the example, the calculator, inequalities and the find- k example, then the three forms, then Vieta if reached.

THE PICTURE



$\Delta > 0$: two roots, the curve crosses the x -axis twice $\Delta = 0$: one repeated root (two equal roots), the curve touches the x -axis
 $\Delta < 0$: no real roots, the curve never reaches it

What the factorised form gives you for free: the top of the flight is halfway between kick and landing. The formula $x = -b/(2a)$ gives the same line, and still works when there are no roots to average.

THREE FORMS, THREE FACTS

Form	Formula	What it hands you	In the example
Traditional (standard)	$y = ax^2 + bx + c$	Crosses the y -axis at c . The discriminant $\Delta = b^2 - 4ac$ counts the roots.	$2x^2 - 12x + 10$: y -intercept 10; $\Delta = 144 - 80 = 64 > 0$, so two roots.
Factorised	$y = a(x - r_1)(x - r_2)$	Crosses the x -axis at the roots r_1, r_2 . The vertex sits halfway between them.	$2(x - 1)(x - 5)$: roots 1 and 5; axis of symmetry $x = 3$.
Vertex	$y = a(x - h)^2 + k$	Vertex at (h, k) : a minimum if $a > 0$ (concave up), a maximum if $a < 0$ (concave down).	$2(x - 3)^2 - 8$: minimum value -8 , at $x = 3$.

INEQUALITIES, AND THE EXAM QUESTION

$2x^2 - 12x + 10 > 0$: positive outside the roots, $x < 1$ or $x > 5$, written $]-\infty, 1[\cup]5, +\infty[$. For ≤ 0 : $x \in [1, 5]$. Read the sign off the picture.

The exam question: $f(x) = 2x^2 - 4x + k$, find k so that $f(x) > 0$ for all x . Two conditions: opens upward ($a = 2$, given) and never touches the axis ($\Delta = 16 - 8k < 0$), so $k > 2$. With ≥ 0 the touch is allowed, $\Delta \leq 0$, so $k \geq 2$.

TRAPS

- The minus sign in $x = -b/(2a)$. Drop it and the vertex lands on the wrong side.
- " > 0 " versus " ≥ 0 ". The equals sign allows $\Delta = 0$, one touching root.
- " $f(x) > 0$ for all x " needs two things: $\Delta < 0$ and $a > 0$. In the notes' example $a = 2$ is given, so only Δ matters.
- If $c = 0$, one root is 0: factor out x , as in $3x^2 + 12x = 3x(x + 4)$. No formula needed.

LISTEN FOR THESE THREE MOMENTS

1. The discriminant goes on the board and the three cases are sorted. Draw the three pictures next to the three signs.
2. The vertex is found. Does he use $-b/(2a)$ or the midpoint of the roots? They agree.
3. The vertex form appears. He will show both completing the square and plugging in the vertex. Notice which one he says to prefer.

Questions to ask in class

Ask at least one. The first is safe, the second shows you followed, the third is the one you actually want answered. Write down what was said; it is the part you will remember.

1. *[safe]*

If $\Delta < 0$ there are no roots to take the midpoint of. Is the vertex still at $x = -b/(2a)$?

What you expect: Yes. The symmetry does not depend on the roots existing. It gets the symmetry said out loud, which is the idea the whole section rests on.

What he said:

2. *[shows you followed]*

In the example where we find k , $a = 2$ is fixed. If a were unknown as well, would " $f(x) > 0$ for all x " need $a > 0$ in addition to $\Delta < 0$?

What you expect: Yes. The notes' example never has to say so because a is given. This is the version the exam likes. Ask it right after part (e) of the find- k example.

What he said:

3. *[the one you want answered]*

When the calculator gives the vertex as a decimal, how do we write the vertex form exactly, and does the paper want the fraction?

What you expect: Example 4 gives $V(-2.5, 60.75)$ straight off the calculator. Expect: use $x = -b/(2a)$ as a fraction, exact form on paper, decimals only when the question asks for them.

What he said:

Later. For the HL hour on Wednesday 14 October: do Vieta's formulas still hold when the roots are complex? If the sum and product are real, is that another way to see why the roots come in a conjugate pair? (Yes to both. His first answer will probably be the quadratic formula, $-b/(2a) \pm i\sqrt{-\Delta}/(2a)$.)

Calculator. Casio: Equation, Polynomial degree 2 for roots. Graph mode, G-Solv (F5): ROOT, MAX or MIN for the vertex, Y-ICPT.

Next. Thursday's double continues 2.2 (examples 4 and 5). No HL hour this Wednesday, fall break. HL returns the week of 12 October with 2.12, sum and product of roots (Vieta): $r_1 + r_2 = -b/a$, $r_1 r_2 = c/a$, then the same idea for any polynomial. Section 2.3, functions, domain and range, also starts that week. Exam II is Thursday 29 October and covers 2.1 to 2.5.